

Group Office Hours

10/02/21

ECON 301
Spring 2021



$$f(x; \alpha, \beta) = \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)} 1_{[0, \infty)}(x)$$

$$\alpha = 2$$

$$\beta = \lambda = 0.7$$

$$f(x; 2, 0.7) = \frac{0.7^2 x e^{-0.7x}}{\Gamma(2)} 1_{[0, \infty)}(x)$$

If f is a p.d.f. $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\int_{-\infty}^{\infty} \frac{0.7^2 x^{2-1} e^{-0.7x}}{\Gamma(2)} 1_{[0, \infty)}(x) dx = \int_0^{\infty} \frac{0.7^2 x e^{-0.7x}}{\Gamma(2)} dx = 1$$

$$E[X] = \int_0^{\infty} x \left(\frac{0.7^2 x e^{-0.7x}}{\Gamma(2)} \right) dx = \int_0^{\infty} \frac{0.7^2 x^2 e^{-0.7x}}{\Gamma(2)} dx \quad \Gamma(2) = (2-1)! = 1$$

$$= \int_0^{\infty} 0.7^2 x^2 e^{-0.7x} dx$$

$$= 0.7^2 \int_0^{\infty} x^2 e^{-0.7x} dx$$

$$= \frac{\Gamma(3)}{0.7} \frac{0.7}{\Gamma(3)} \int_0^{\infty} 0.7^2 x^{3-1} e^{-0.7x} dx \quad \alpha' = 3, \beta = 0.7$$

$$= \frac{\Gamma(3)}{0.7} \underbrace{\int_0^{\infty} \frac{0.7^3 x^{3-1} e^{-0.7x}}{\Gamma(3)} dx}_{=1} = \frac{\Gamma(3)}{0.7} = \frac{2!}{0.7} = \frac{2}{0.7}$$

For Task 1:

$$E[Y|X=x] = \frac{x}{2}$$

$$E^*(Y|X) = \frac{x}{2}$$

$$\alpha = 0$$

$$\beta = 1/2$$

$$E^*(Y|X) = \alpha + \beta X$$

$$\alpha = E(Y) - \beta E(X)$$

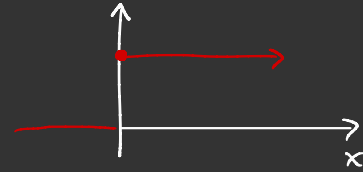
$$\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$Y|X \sim \frac{1}{x} \mathbb{1}_{[0, x]}(y)$$



$$X \sim \frac{0.7^2 x e^{-0.7x}}{\Gamma(2)} \mathbb{1}_{[0, \infty)}(x)$$

$$0 \leq x < \infty$$



Bayes' Rule: $P(A|B) = \frac{P(A \cap B)}{P(B)}$

$$\Rightarrow P(A|B) P(B) = P(A \cap B)$$

$$f(x, y) = f(y|x) \cdot f(x) = \left(\frac{1}{\cancel{x}} \mathbb{1}_{[0, x]}(y) \right) \left(\frac{\cancel{0.7^2 x} e^{-0.7x}}{\cancel{\Gamma(2)}} \mathbb{1}_{[0, \infty)}(x) \right)$$

1

$$f(x, y) = 0.7^2 e^{-0.7x} 1_{[0, x]}(y) 1_{[0, \infty)}(x)$$

$$f_x(x) = \int_{-\infty}^{\infty} 0.7^2 e^{-0.7x} 1_{[0, x]}(y) 1_{[0, \infty)}(x) dy$$

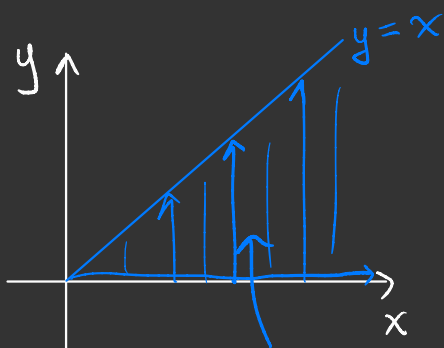
$$= 0.7^2 e^{-0.7x} 1_{[0, \infty)}(x) \int_{-\infty}^{\infty} 1_{[0, x]}(y) dy$$

$$= 0.7^2 e^{-0.7x} 1_{[0, \infty)}(x) \underbrace{\int_0^x 1 dy}_{\downarrow x}$$

$$= 0.7^2 e^{-0.7x} 1_{[0, \infty)}(x) \left(y \Big|_0^x \right) = 0.7^2 e^{-0.7x} x 1_{[0, \infty)}(x)$$

$$f(x, y) = 0.7^2 e^{-\lambda x} 1_{[0, x]}(y) 1_{[0, \infty)}(x)$$

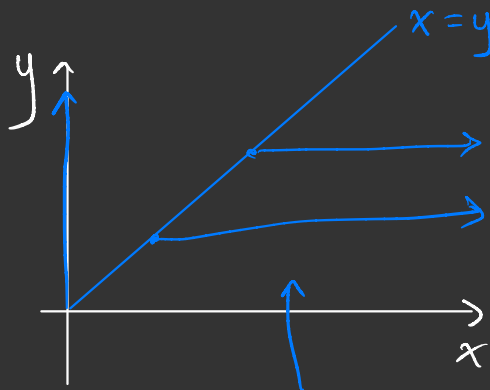
$$f_y(y) = \int_{-\infty}^{\infty} 0.7^2 e^{-\lambda x} 1_{[0, x]}(y) 1_{[0, \infty)}(x) dx$$



$$\{(x, y) \mid 0 \leq x < \infty, 0 \leq y \leq x\}$$

$$\downarrow$$

$$1_{[0, \infty)}(x) \cdot 1_{[0, x]}(y)$$



$$\{(y, x) \mid 0 \leq y < \infty, y \leq x < \infty\}$$

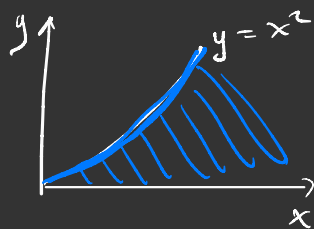
$$1_{[0, \infty)}(y) \cdot 1_{[y, \infty)}(x)$$

$$\int_{-\infty}^{\infty} 0.7^2 e^{-\lambda x} 1_{[0, \infty)}(y) \cdot 1_{[y, \infty)}(x) dx$$

$$= \int_y^{\infty} 0.7^2 e^{-\lambda x} 1_{[0, \infty)}(y) dx$$

$$0 \leq x < \infty$$

$$0 \leq y \leq x^2$$



$$\{(x, y) \mid 0 \leq x < \infty, 0 \leq y \leq x^2\}$$

$$1_{[0, \infty)}(x) \cdot 1_{[0, x^2]}(y)$$

=

$$\{(y, x) \mid 0 \leq y < \infty, y^2 \leq x < \infty\}$$

$$1_{[0, \infty)}(y) \cdot 1_{[y^2, \infty)}(x)$$

$$E[X^m Y^n] \neq E[X^m] E[Y^n]$$

↖ is computed
using the
joint p.d.f.

Task 5

$$E^*(Y|X) = \alpha + \beta X$$

$$\alpha = E(Y) - \beta E(X)$$

$$\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$E^*(Z|X) = \alpha' + \beta' X$$

$$\alpha' = E(Z) - \beta' E(X)$$

$$\beta' = \frac{\text{Cov}(Z, X)}{\text{Var}(X)}$$